

Semantic Knowledge Representation and Reasoning:

A First-Order Logic Approach for Explainable AI

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Research Questions

- Q1.** How can semantic parsing improve explainability and auditability of AI systems by translating natural language into verifiable first-order logic (FOL) representations?
- Q2.** How can rule-based parsing ensure accurate and interpretable FOL translations?
- Q3.** How does symbolic reasoning enhance traceability compared to black-box models?

Overview

This project develops a symbolic reasoning system that converts natural language into first-order logic (FOL) to improve AI explainability. By combining rule-based parsing with logical inference, the system enables transparent, verifiable, and auditable reasoning—addressing the “black box” problem of current AI models.

Applications include domains such as **healthcare, finance, and law**, where explainability and compliance are critical.

Methodology

- **NL-to-FOL Parser:** Rule-based system using linguistic parsing tools (e.g., NLTK, spaCy).
- **FOL Reasoning Engine:** Performs automated inference using libraries like Kanren or pyDatalog.
- **Explainability Module:** Generates human-readable proof traces and visualizations of inference steps.
- **Baselines:** Compared with LLM-based parsing and reasoning systems (LLM-Parser, LLM-Reasoner).

Expected Deliverables

1. Functional **NL-to-FOL parser** and **reasoning engine**.
2. **Proof visualization** and **explanation module** for explainable inference.

3. **Evaluation results** on benchmark datasets such as FOLIO and LogicNLI.
4. **Comparative analysis** showing improved interpretability over LLM baselines.

References

- IJCAI (2020) – *Towards Trustable Explainable AI*.
- arXiv (2024) – *Logic-Based Explainability: Past, Present & Future*.
- AAMAS (2021) – *Better Metrics for Evaluating Explainable AI*.
- EMNLP (2021) – *LogicNLI: Diagnosing FOL Reasoning Ability*.
- Yale-LILY (2022) – *FOLIO: Natural Language Reasoning with FOL*.